

Adversarial Exploration for Detecting Emergent Stratification in LLM Decision Making

Joaquin Parra

Context

Bai et al. (2025) provide an account of the exploration-exploitation dilemma centered on stereotype formation in sequential decision-making environments. In this sense, social beliefs may not emerge exclusively from pre-existing associations, but from constrained sampling: when early observations presume to favor a particular group-role pairing, decision-makers may exploit that apparent regularity, thus reducing exploration of alternative options (see Bai, Fiske & Griffiths, 2022). Over time, these initial observations can evolve to structured impressions about entire groups. Recent work extends this framework to large language models (LLMs), showing their capacity to develop novel social biases about artificial demographic groups: “LLMs demonstrate a new kind of bias —the creation of novel stereotypes— which manifests over repeated interactions in stateful frameworks” (Bai et al., 2025, p. 10). Most surprisingly, they report that LLMs appear to be “even more prone than humans to develop such biases, even when underlying differences do not exist” (Bai et al., 2025, p. 10). These biases manifest as stratified task allocations, which are more prominent in newer frontier models despite their promise for impartiality; effect presumably caused by higher optimizing power towards given objectives. This suggests that LLMs may not just reproduce inherited social biases, but also generate new ones through adaptive interaction.

Summary of the Proposal

The proposed research investigates whether adversarial debiasing can be used not only to reduce bias after it appears, but also to detect signs of stratification as they emerge. In conventional adversarial debiasing, a predictive model is trained to perform a task while an adversary model attempts to recover a sensitive attribute from the model’s representations or outputs. If the adversary cannot successfully identify the sensitive attribute, this is taken to indicate that group-related information is less available to the model’s decision process. In this setting, the main question becomes whether the model begins to make artificial group identity easier to infer as it gains experience.

The proposed framework therefore introduces an adversary into a sequential allocation task. An LLM assigns candidates from artificial groups to job categories across repeated rounds, having all groups equally qualified in the underlying environment. Periodically, an adversarial model will observe the LLM decisions and explanations with explicit group labels removed and attempt to infer the candidate’s group identity. If the adversary can infer group identity from the LLM’s decisions or explanations, this would suggest that the model has started to feature differentiated treatment towards groups. In this case, group identity would have become embedded in the model’s behavior despite the task environment having no real group differences.

The adversary's predictions can be used to guide the model's further exploration. Rather than relying on a fixed instruction to avoid bias, the system would monitor whether the LLM's allocation history make artificial group identity inferable. When this occurs, it would be an indication that the model is beginning to form group–role associations. The intervention would then direct the model toward group–role pairings that have been insufficiently considered, thus incentivizing exploration. Therefore, this framework serves both as a diagnostic tool for emerging stratification and as a mechanism for redirecting the model before early patterns become stable biases.

Relevance

In terms of relevance, this study connects adversarial debiasing to recent work on dynamic bias formation in LLMs. While prior work has been focused on detecting or reducing biases already present in training data (see also Yang et al. & Zhang et al.), the exploration based account discussed above suggests that bias can also emerge through interaction, even when no real group differences exist. In LLM decision-making, this means that repeated choices and feedback may eventually produce group–role associations that were not originally present.

The proposed study extends this work by asking whether this empirical stratification can be detected while still in its formation phase. Existing metrics such as the ones presented in the context study (SI, BDG and GASI) describe the structure and stability of allocation patterns after they disappear. Therefore, the proposed adversarial model adds a complementary signal by testing whether group identity can be inferred from the model's decisions and explanations.

As we enter the so-called agentic era, LLMs become autonomous and become useful in systems that remember past interactions, receive feedback, and make repeated recommendations and allocations. In these settings, fairness risks may arise not only from isolated outputs, but from the way models form beliefs over time. In this sense the proposed adversarial exploration structure offers a way to detect and redirect those dynamics before early patterns become increasingly stable stereotypes.

References

- Bai, X., Fiske, S. T., & Griffiths, T. L. (2022). Globally Inaccurate Stereotypes Can Result From Locally Adaptive Exploration. *Psychological science*, 33(5), 671–684. <https://doi.org/10.1177/09567976211045929>
- X. Bai, A. Wang, I. Sucholutsky, & T. L. Griffiths, Explicitly unbiased large language models still form biased associations, *Proc. Natl. Acad. Sci. U.S.A.* 122 (8) e2416228122, <https://doi.org/10.1073/pnas.2416228122> (2025).
- Wu, A. J., Liu, R., Bai, X., & Griffiths, T. L. (2026). Large language models develop novel social biases through adaptive exploration. *arXiv*. <https://doi.org/10.48550/arXiv.2511.06148>
- Yang, J., Soltan, A.A.S., Eyre, D.W. et al. An adversarial training framework for mitigating algorithmic biases in clinical machine learning. *npj Digit. Med.* 6, 55 (2023). <https://doi.org/10.1038/s41746-023-00805-y>

Zhang, B. H., Lemoine, B., & Mitchell, M. (2018). Mitigating unwanted biases with adversarial learning. In Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society (pp. 335–340). Association for Computing Machinery. <https://doi.org/10.1145/3278721.3278779>